

fundamentals of logic

1. Basic Connectives and Truth Table

statements (propositions): declarative sentences that are either true or false
.. but Not both

ex: Kanpur is in india

or $2 \leq 15$

It can not be a question

It also can't contain variables

$x \leq 9$ is not a proposition

→ primitive and Compound statements

Combined from primitive statements by logical operators
Connectives or by negation

logical Connectives:

1. $P \wedge Q$ Conjunction (AND)
2. $P \vee Q$ Disjunction (OR)
3. $P \rightarrow Q$ If P then Q
4. $P \leftrightarrow Q$ P iff Q (Bidirectional)

P	Q	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	0	0
1	1	1	1	1	1

S: Mohan goes out for a walk
 t: the moon is out
 U: It's snowing

$$(t \wedge \neg U) \rightarrow S$$

: If the moon is out and it's not snowing then Mohan goes out for a walk.

→ A compound statement is called a tautology if it's true for all truth value assignments for its component statement.

If a compound statement is false for all such assignments. Then it's called a contradiction.

$$p \rightarrow (p \vee q) : \text{tautology}$$

$$p \wedge (\neg p \wedge q) : \text{contradiction}$$

Note: $p \rightarrow q = \neg p \vee q$.

Laws of logic:-

(AND, OR, NOT) is a Complete Set

$$\left. \begin{aligned} p \rightarrow q &\Leftrightarrow \neg p \vee q \\ p \leftrightarrow q &\Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p) \\ &\Leftrightarrow (\neg p \vee q) \wedge (\neg q \vee p) \\ p \oplus q &\Leftrightarrow (p \vee q) \wedge \neg(p \wedge q) \\ &\Leftrightarrow (p \vee q) \wedge (\neg p \vee \neg q) \end{aligned} \right\}$$

Let S be a statement. If S contains no logical connectives other than $\wedge$ and $\vee$, then the dual of S , denoted by S^d , is the statement obtained from S by replacing each occurrence of $\wedge$ and $\vee$ by $\vee$ and $\wedge$ respectively and each occurrence of T_0 and F_0 by F_0 and T_0 respectively.

$$S: (p \wedge \neg q) \vee (\neg p \wedge T_0)$$

$$S^d: (p \vee \neg q) \wedge (\neg p \vee F_0)$$

dual of $p \rightarrow q$ is $(\neg p \vee q)^d \Leftrightarrow \neg p \wedge q$

Principle of Duality :-

Let S and T be statements
if $S \Leftrightarrow T$ then $S^d \Leftrightarrow T^d$

Simplification of Compound Statement

$$(p \vee q) \wedge \neg(\neg p \wedge q)$$

$$\Leftrightarrow (p \vee q) \wedge (p \vee \neg q)$$

$$\Leftrightarrow p \vee (q \wedge \neg q)$$

$$\Leftrightarrow p \vee F_0 \Leftrightarrow p.$$

Rules of Inference:-

an argument: $\underbrace{(p_1 \wedge p_2 \wedge \dots \wedge p_n)}_{\text{premises}} \rightarrow \underbrace{q}_{\text{conclusion}}$

is a valid argument

$\longleftrightarrow (p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology.

Ex: Statements:

p: Roger studies

a: Roger plays Tennis

2: Roger passes discrete mathematics

Premise:

p_1 : if Roger studies then he will pass discrete math.

p2: if Roger does not play tennis
then he will study

P_3 : Roger failed discrete math

Determine whether the argument.

$$(p_1 \wedge p_2 \wedge p_3) \rightarrow q$$
$$p_1: p \rightarrow r \quad p_2: \neg q \rightarrow p \quad p_3: \neg r$$
$$\therefore (p_1 \wedge p_2 \wedge p_3) \rightarrow q \Leftrightarrow$$
$$[(p \rightarrow q) \wedge (\neg q \rightarrow p) \wedge \neg r] \rightarrow q$$

→ Tautology
(5)

Rules of inference related to the language of propositions

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

Modus Ponens (Method of affirming)

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Law of Syllogism

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

$$\begin{array}{l} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

Modus Tollens (Method of denying)

$$\begin{array}{l} p \rightarrow q \\ q \\ \hline \therefore p \end{array}$$

fallacy.

$$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$$

Addition.

$$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$$

Simplification.

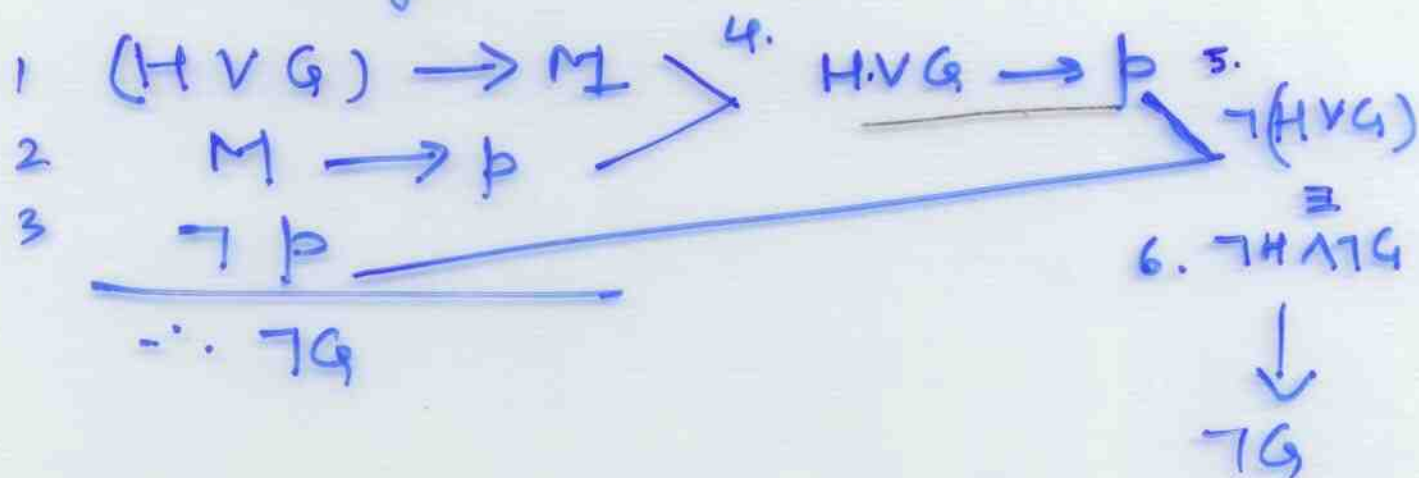
If Horses fly or Cows eat grains.
 then the mosquito is the national bird
 if the mosquito is the national bird
 then peanut butter tastes good on
 hot dogs. but peanut tastes terrible
 on hot dogs. therefore cows does not
 eat grains.

Horses fly H

Cows eat grains G

Mosquito is N.B M

Peanut butter
 tastes good on P ✓
 hot dogs.



Use of Quantifiers:-

A declarative sentence is an open statement if

- (1) It contains one or more variables and.
- (2) It's not a statement, but
- (3) It becomes a statement when the variables in it are replaced by certain allowable choices

notations:

$p(x)$: the number $x+2$ is an even integer

$q(x,y)$: the numbers $y+2$, $x-y$ and $x+2y$ are even integers

$p(5)$: false $\neg p(7)$: true

$q(4,2)$: True $p(6)$: True $\neg p(8)$: false

$q(3,4)$: False

therefore:

for some x , $p(x)$ is true

for some x , $\neg p(x)$ is true.

for some x, y $q(x, y)$ is true

for some x, y $\neg q(x, y)$ is true

Existential quantifier:

for some x : $\exists x$

Universal quantifier:

for all x : $\forall x$

x in $p(x)$: free variable

x in $\exists x p(x)$: bound variable

$\rightarrow \exists x p(x)$ is either
true or false

$$p(x): x \geq 0$$

$$q(x): x^2 \geq 0$$

$$r(x): x^2 - 3x - 4 \leq 0$$

$$s(x): x^2 - 3 > 0$$

$$\exists x [p(x) \wedge r(x)]: \text{True } x=4$$

$$\forall x [p(x) \rightarrow q(x)]: \text{True}$$

$$\exists x [p(x) \rightarrow q(x)]: \text{True}$$

$$\forall x [q(x) \rightarrow r(x)]: \text{false } x=1$$

$$\forall x [r(x) \vee s(x)]: \text{false } x=5/2$$

Translating from English into predicate logic

Ex:- "Every student in this class has studied calculus"

1. first we rewrite the statement so that we can easily identify the appropriate Quantifiers to use.

"for every student, if the student is in this class, then that student has studied calculus"

2. Next, introduce a variable s to denote a student

"for every student s, if s is in this class, then s has studied calculus"

3. Define the predicates
 $\text{in_class}(s) = "s \text{ is in this class}"$
 $C(s) = "s \text{ has studied calculus}"$

4. $\forall s (\text{in_class}(s) \rightarrow C(s))$.

Ex:- "Every bird can fly"

1. We can reformulate this sentence as follows.

"for every b , if b is a bird then b can fly"

$\forall b$ $\text{is_bird}(b)$ $\text{can_fly}(b)$

$\forall b (\text{is_bird}(b) \rightarrow \text{can_fly}(b))$

Note: if we specify that b is a variable that ranges over birds.

Then $\forall b \text{ can_fly}(b)$

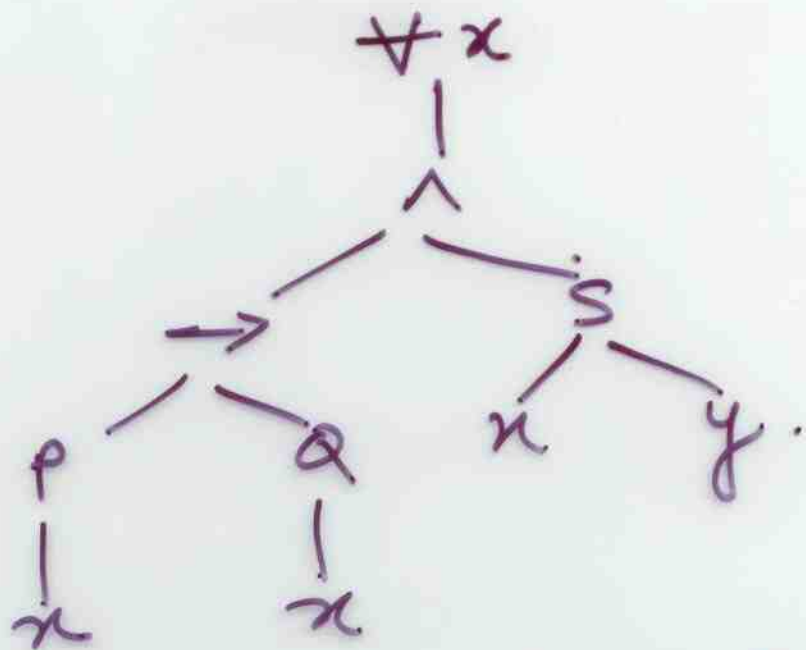
Ex:- "Nobody in this class ^{can} speak Russian or French"

There is no person x such that x is in this class and x can speak Russian and x can speak French.

$\neg \exists x \cdot (\text{in_class}(x) \wedge \text{speaks}(x, \text{Russian}) \wedge \text{speaks}(x, \text{French}))$

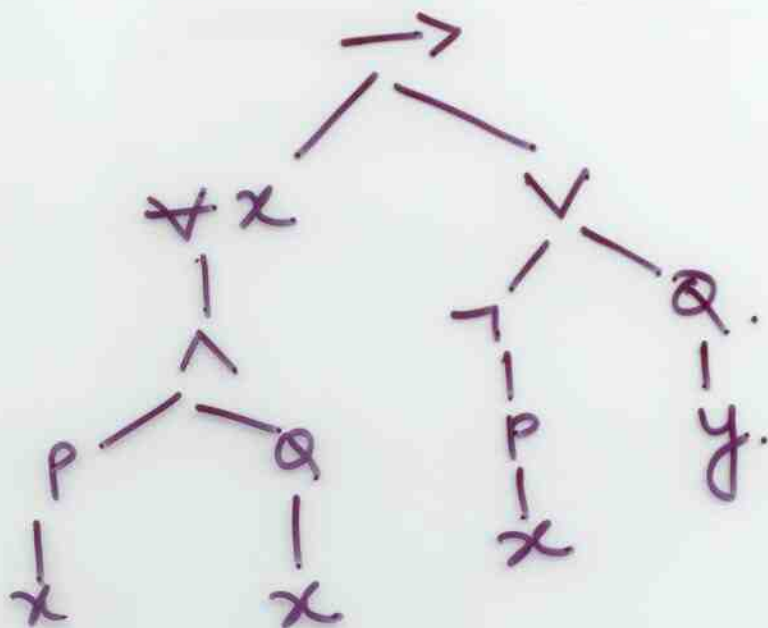
Predicate logic as a formal language.

$$\forall x((P(x) \rightarrow Q(x)) \wedge S(x, y))$$



The quantifier $\forall x$ refers to all occurrences of x below it in the syntax tree. We say that the quantifier binds all the variable occurrence. A variable that is not bound by any quantifier is called free. e.g. variable y is free in the formula above.

$$(\neg \forall x (P(x) \wedge Q(x))) \rightarrow (\neg P(x) \vee Q(y))$$



occurrence of x in
 $\neg P(x)$ is free.

Approach in a predicate logic Proof.

- Remove the Quantifiers
- manipulate the Unquantified wff's
- Re-introduce the Quantifiers.

Name	Abbr	Given	Can Infer	Restrictions
Universal Instantiation	UI	$\forall x P(x)$	$P(x) \text{ or } P(a)$	
Existential Instantiation	EI	$\exists x P(x)$	$P(y)$	only on first introduction of variable y
Universal generalisation	UG	$P(x)$	$\forall x P(x)$	x was not introduced by EI or by a hypothesis
Existential generalisation	EG	$P(x)$ or $P(a)$	$\exists x P(x)$	

UI

All humans are mortal. Socrates is human
Therefore Socrates is mortal

$H(x) \equiv$ "x is human"

$M(x) \equiv$ "x is mortal"

✓ $\forall x (H(x) \rightarrow M(x))$

$H(s)$

$H(s) \rightarrow M(s)$

$M(s)$

hypothesis.
hypothesis

UI

mp.

EI

Prove:

$$\underline{(\neg \forall x)(P(x) \rightarrow Q(x)) \wedge (\exists x)P(x) \rightarrow (\exists x)Q(x)}$$

1.	$\neg \forall x (P(x) \rightarrow Q(x))$	H
2.	$(\exists x) P(x)$	H
3.	$P(y)$	2, EI
4.	$P(y) \rightarrow Q(y)$	1, UI
5.	$Q(y)$	3, 4, mp.
6.	$\exists x Q(x)$	5, EG

Not: We can not reverse step 3 and 4 here
if UI names an arbitrary element y , then EI can not be used to deduce anything about y .

UG

$$(\forall x)(P(x) \rightarrow Q(x)) \wedge (\forall x)P(x) \rightarrow (\forall x)Q(x)$$

- | | | |
|----|--------------------------------------|-------------|
| 1. | $(\forall x)(P(x) \rightarrow Q(x))$ | H |
| 2. | $\forall x P(x)$ | H |
| 3. | $P(y) \rightarrow Q(y)$ | $\perp, UI$ |
| 4. | $P(y)$ | 2, UI |
| 5. | $Q(y)$ | 4, 3, mp |
| 6. | $(\forall x)Q(x)$ | 5, UG |

EG

$$(\forall x)P(x) \rightarrow (\exists x)P(x)$$

- | | | |
|----|-------------------|-------|
| 1. | $(\forall x)P(x)$ | H |
| 2. | $P(x)$ | 1, UI |
| 3. | $(\exists x)P(x)$ | 2, EG |

Bogus Proof.

$$\textcircled{1} \quad \left. \begin{array}{l} (\exists x) (P(x) \rightarrow Q(x)) \\ (\exists y) P(y) \end{array} \right\} \Rightarrow (\exists w) Q(w)$$

1.	$(\exists x) (P(x) \rightarrow Q(x))$	Given	
2.	$(\exists y) P(y)$	Given	
3.	$P(u)$	2, EI	
4.	$P(u) \rightarrow Q(u)$	1, EI	X
5.	$Q(u)$	3, 4, mp	
6.	$(\exists w) Q(w)$	5, EG	

step 4 not allowed.
u has already been
used.

$$\textcircled{2} \quad (\exists x) P(x) \Rightarrow (\forall x) P(x)$$

1.	$(\exists x) P(x)$	Given	
2.	$P(u)$	1, EI	
3.	$\forall x P(x)$	2, VG	X

can not generalised from an
element introduced by EI

logical relationships with quantifiers

Law

$$\left. \begin{aligned} \neg \forall x P(x) &\equiv \exists x \neg P(x) \\ \neg \exists x P(x) &\equiv \forall x \neg P(x) \end{aligned} \right\} \text{De-Morgan.}$$

$$\forall x P(x) \wedge \forall x Q(x) \equiv \forall x (P(x) \wedge Q(x)).$$

distributivity of $\forall$ over $\wedge$

✓

$$\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x).$$

distributivity of $\exists$ over $\vee$

✓